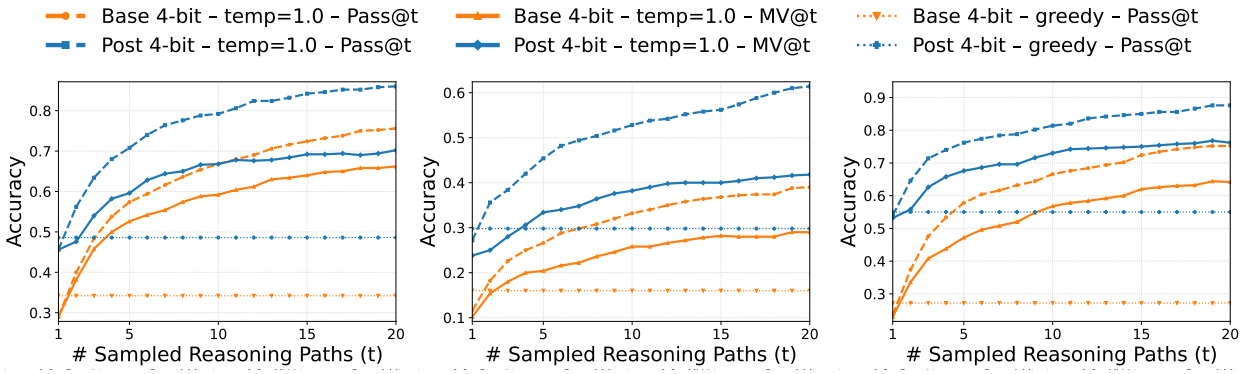

(a) Llama-3B on MATHQA **(b)** Qwen-2B on MATH **(c)** Qwen-2B on GSM8K

Figure 3 Post-training self-consistency improves sampling accuracy. Dashed: Pass@t (oracle upper bound), solid: MV@t (majority over t samples), dotted: greedy ($\tau = 0$) accuracy. (Blue): post-trained model. (Orange): base model. Curves computed over 500 prompts.

Table 3 Post-training self-consistency improves performance across general reasoning benchmarks. Models trained on datasets (columns) tested on benchmarks (rows). Bottom rows show generalization to unseen benchmarks: SVAMP (math), GPQA (science), CSQA (commonsense). All = joint training on combined datasets. Arrows show absolute gains over instruction-tuned model.

Test	Qwen2B (post-trained on)					Llama3B (post-trained on)				
	Base	MATH	GSM	MQA	All	Base	MATH	GSM	MQA	All
MATH	10.4	↑10.0	↑3.8	↑10.8	↑12.2	32.0	↑19.4	↑16.4	↑18.2	↑21.2
GSM	27.0	↑20.0	↑25.6	↑22.6	↑27.8	69.6	↑6.0	↑6.8	↑8.4	↑10.8
MQA	7.4	↑12.6	↑17.0	↑15.4	↑21.4	24.6	↑14.0	↑13.4	↑21.2	↑21.6
SVAMP	48.3	↑19.0	↑18.0	↑17.0	↑27.7	71.3	↑6.0	↑6.4	↑9.7	↑7.1
GPQA	0.5	↑6.0	↑5.3	↑12.8	↑16.3	0.7	↑5.4	↑6.3	↑9.8	↑10.7
CSQA	3.8	↑19.8	↑43.0	↑54.0	↑59.6	53.0	↑7.4	↑10.6	↑11.6	↑11.0

In multi-agent debate, we evaluate the agreement metric $d_M^{\theta,\tau}(x)$ defined in Sec. 3. Base small LMs initially struggle to reach meaningful consensus: for Qwen-2B on GSM8K, most of the "consensus" comes from random tie-breaking (1/3 agreement) or weak majorities (2/3), with only 13.4% reaching unanimity (Fig. 12, App. O). Post-training with MACA thoroughly addresses this: non-parseable responses drop from 13.8% to 0.6%, no-agreement cases from 45.6% to 19.8%, while unanimous agreement triples from 13.4% to 43.4%. This confirms that MACA improves both individual reasoning quality and collaborative grounding, enabling genuine consensus rather than noisy aggregation (agreement distributions in App. O).

Impact on inference-time sampling. We next examine how gains in self-consistency translate to inference-time performance under various sampling regimes. For each prompt, we draw t trajectories and report two metrics: Pass@t, the fraction of prompts for which at least one of the first t samples is correct (Chen et al., 2021), and MV@t, the fraction for which a majority vote over those t samples matches the ground truth (ties counted as incorrect) (Li et al., 2024). As shown in Fig. 3, post-training (blue) (i) lifts greedy (τ=0) accuracy, (ii) increases MV@t at each fixed t, and (iii) raises Pass@t, the empirical sampling upper bound, indicating higher achievable accuracy at any given sampling budget. When additional inference compute is available (i.e., $t > 1$), sampling techniques continue to see gains on top of the post-training improvements, showing MACA is complementary to inference-time sampling. The same pattern holds with full-precision, with improvements comparable to their 4-bit counterparts (App. N). Finally, self-consistency gains strongly correlate with accuracy improvements ($r > 0.86$ across all tested inference conditions; see App. I).

Improving self-consistency on math datasets improves generalization to unseen benchmarks. These consistency and accuracy gains extend beyond the training distribution: training to improve internal self-consistency on any mathematical dataset enhances performance across diverse reasoning tasks, including previously unseen domains. Building on recent work showing math training enhances general reasoning (DeepSeek-AI et al., 2026; Akter et al.,

Table 4 Multi-agent: Post-training with debate (MV) is comparable to ground-truth (GT).

Llama-8B	Dataset	Debate	SFT		KTO		DPO		GRPO	
			GT	*MV*	*GT*	*MV*	*GT*	*MV*	*GT*	*MV*
Single-Agent	MATH	22.93	**23.73**	23.00	**41.20**	39.40	45.13	**46.40**	29.07	**31.13**
	GSM8K	57.93	40.20	**41.67**	**72.60**	70.87	76.33	**77.67**	61.27	**66.87**
Multi-Agent	MATH	32.80	**34.00**	33.40	**56.00**	55.80	**61.80**	60.80	**48.60**	44.80
	GSM8K	74.00	**65.20**	**65.20**	**81.20**	79.40	81.60	**83.0**	83.20	**84.20**

Table 5 Single-agent: RL on debate majority-vote with peer context (MV-DPO) outperforms RL on single-round majority vote (TTRL, ScPO). Arrows show gains over instruction-tuned model. Statistics gathered over 3 seeds.

	MATH				GSM8K			
	Qwen-2B	Llama-3B	Phi-4B	Llama-8B	Qwen-2B	Llama-3B	Phi-4B	Llama-8B
TTRL	↑**18.0**$_{\pm 2.9}$	↑5.3$_{\pm 5.7}$	↑6.1$_{\pm 2.1}$	↑7.5$_{\pm 0.2}$	↑**24.5**$_{\pm 1.6}$	↓22.3$_{\pm 8.3}$	↑4.8$_{\pm 1.6}$	↓13.5$_{\pm 5.7}$
ScPO	↑2.3$_{\pm 1.1}$	↑3.4$_{\pm 0.2}$	↑0.1$_{\pm 0.5}$	↑3.7$_{\pm 0.6}$	↑15.9$_{\pm 3.4}$	↑1.6$_{\pm 0.8}$	↑5.9$_{\pm 0.5}$	↑12.9$_{\pm 1.2}$
MV-DPO	↑16.7$_{\pm 0.4}$	↑**12.5**$_{\pm 0.7}$	↑**6.9**$_{\pm 0.2}$	↑**17.1**$_{\pm 0.8}$	↑23.8$_{\pm 0.9}$	↑**3.7**$_{\pm 0.3}$	↑**7.3**$_{\pm 0.3}$	↑**18.2**$_{\pm 0.2}$

2025), we demonstrate that improving self-consistency—internal consensus strength—on math is essential for reliably unlocking these capabilities. Table 3 reports results for models trained on MATH, GSM8K, or MathQA individually, and on all three combined (All), using MV-DPO. Training on any single dataset improves performance across all reasoning tasks, including unseen math (SVAMP), science (GPQA), and commonsense reasoning (CSQA). Joint training achieves further improvements across nearly every benchmark, demonstrating that diverse training data amplifies self-consistency gains.

5.3 Ablation Study

To understand the sources of performance gains from MACA, we conduct ablation studies examining key components. We show that self-generated consensus signals outperform ground-truth supervision, including peer context during training improves relative grounding and debate utilization, and multi-round debate provides stronger signals than single-round majority vote. We also demonstrate in App. L that self-supervised preference learning serves as an effective implicit format reward by reinforcing more efficient and concise chain-of-thought construction, though most gains stem from fundamental reasoning improvements rather than formatting alone.

RL with debate is comparable to RL with ground-truth. Table 4 compares post-training with debate majority-vote labels, derived from the model's own consensus without external supervision, to ground-truth labels on Llama-3B. Across post-training methods and in both single- and multi-agent settings, using Debate-MV is consistently comparable to ground-truth supervision, a trend that holds across other models as well (App. E). We additionally show in App. J that this comparable performance between Debate-MV and ground-truth supervision extends to general reasoning improvements under MACA across all proposed post-training methods and reasoning benchmarks tested, as observed in Table 3. Because Debate-MV scales naturally with sample size, its parity with ground-truth highlights self-supervised alignment as a promising direction.

RL with debate outperforms RL with single-round majority vote. We evaluate three training conditions in the single-agent setting to understand the impact of training on the debate versus initial round majority vote on the individual agents themselves, as shown in Table 5: (1) initial round majority vote with GRPO (akin to TTRL (Zuo et al., 2025)), (2) initial round majority vote with DPO (akin to ScPO without the weighted loss term (Prasad et al., 2025)), and (3) debate majority vote with DPO including peer context (MV-DPO from MACA). Training on the debate signal improves single-agent performance more than reinforcing the initial round majority vote. Additionally, Table 6 shows that including peer context during preference learning (MV-DPO) improves multi-agent debate performance (more models/datasets in App. F), with the largest gains in final-round individual accuracy, reflecting better use of peer chains-of-thought during deliberation (Fig. 2).

Debate improves consensus quality. Finally, the debate process itself improves consensus signals. Base small LMs produce mostly random or weak majorities initially, but post-training increases unanimous agreement from 27.2%

Table 6 Multi-agent: Post-training with debate context (peer CoTs) outperforms no context. Models: 3B = Llama-3B; 4B = Phi-4B; 8B = Llama-8B. Dataset: GSM8K

	Initial Round Avg		Final Round Avg		Final Round MV	
	NoCtx	Context	NoCtx	Context	NoCtx	Context
3B	65.87	**66.73**	67.87	**70.60**	70.60	**73.80**
4B	74.67	**76.60**	74.60	**77.80**	81.80	**84.40**
8B	75.80	**77.67**	78.00	**81.20**	81.60	**83.00**

to 43.4% between initial and final rounds (Qwen-2B on GSM8K, App. O). This iterative refinement provides more reliable training signals than independent sampling. While majority vote correlates strongly with accuracy (alternative ranking signals like log-probability are explored in App. H), post-training debate creates a self-reinforcing cycle: better debate generates higher-quality signals, producing models that debate more effectively. Iterative training yields continued improvements with diminishing returns, as shown in App. G.

6 Conclusion, Limitations, and Future Work

We introduce self-consistency as an intrinsic property of well-aligned reasoning and present MACA, a self-supervised framework that teaches models to reliably sample coherent reasoning through reinforcement of internal consensus signals. Without external supervision, MACA drives substantial improvements: +27.6% self-consistency on GSM8K, +22.4% Pass@20 on MATH, +23.7% zero-shot accuracy on MATH, strong generalization to unseen benchmarks (+16.3% GPQA, +11.6% CommonsenseQA), and enhanced multi-agent performance (+42.7% on MathQA). Through debate, agents attempt to ground their reasoning in peer context, and we select trajectories where such deliberation aligns with internal consensus. These consensus-aligned examples provide natural supervision for stable reasoning patterns, enabling models to self-improve their reasoning consistency, efficiency, and accuracy without explicit chain-of-thought supervision or external answer verification. Through preference learning on debate signals, models produce more concise chains-of-thought without format rewards and correct cases where aggregation previously led to degeneration, significantly improving the robustness of multi-agent debate. While MACA has limitations, including requiring sufficient base model competence to generate meaningful consensus signals, potentially amplifying existing biases in model outputs, and lacking direct supervision of intermediate reasoning correctness, it demonstrates that consensus reinforcement has robust benefits. Future work could explore alternative consensus methods, confidence-weighted voting, heterogeneous agents, and better leveraging of minority traces. Sizeable gains on difficult unseen tasks (+10.0% on AMC, App. L) suggest consensus alignment unlocks latent capabilities beyond consistency itself. MACA demonstrates that language models can effectively harness internal deliberation for self-alignment, elevating their reasoning capabilities autonomously.